## a Program composition under authenticated actor training exposure

Recorded runs; unequal work. Program occurrences, distinct signatures and reactions are different units.

| Policy | Seen contexts | Unseen contexts | Unseen signatures | Reactions with unseen programs |
|---|---|---|---|---|
| MCTS | 5,192 | 6,288 | 3,890 | 4,952 |
| Greedy I | 863 | 478 | 249 | 467 |
| Greedy II | 866 | 498 | 253 | 475 |

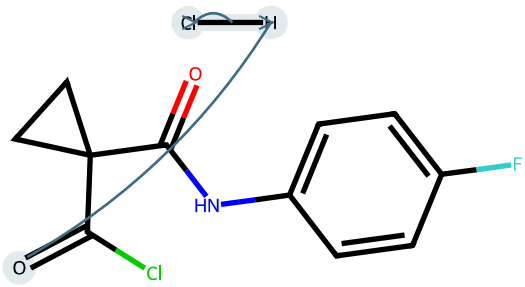
## b Whole-reaction precedent for seen versus unseen MCTS programs

| Composition | Distinct reactions | A/B-supported reactions | Supported targets |
|---|---|---|---|
| Seen | 3,195 | 28 | 24 |
| Training-unseen | 4,952 | 11 | 11 |

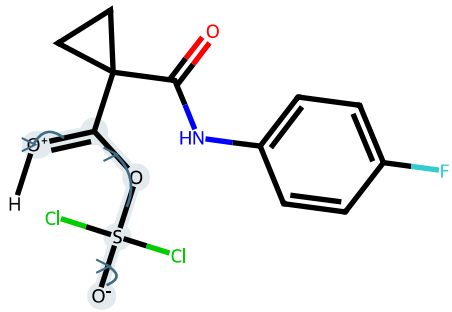
A reaction may have both seen and unseen programs; reaction counts are not additive. Reference C0/C1/C2: Unknown.

## c Selected recorded inverse FLOW actions, not inferred physical intermediates

Action 2 · 2 recorded pair moves




Action 7 · 3 recorded pair moves




Selected complete program: unseen ordinary. All actions retained in linked plates.

Source-backed net chemistry does not establish the exact microscopic electron-arrow order.

Training-unseen electronic composition is not new chemistry, strict-H2 generalization or a reference C1 label.